

DU Capteur et Innovation technologique

Project Presentation

Topic :

Project Progress Presentation Low-Cost Open Flexure Microscope



Presented by :

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Tutors :

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What is an optical microscope ?

- is a scientific instrument
- who uses visible light to enlarge objects of very small dimensions

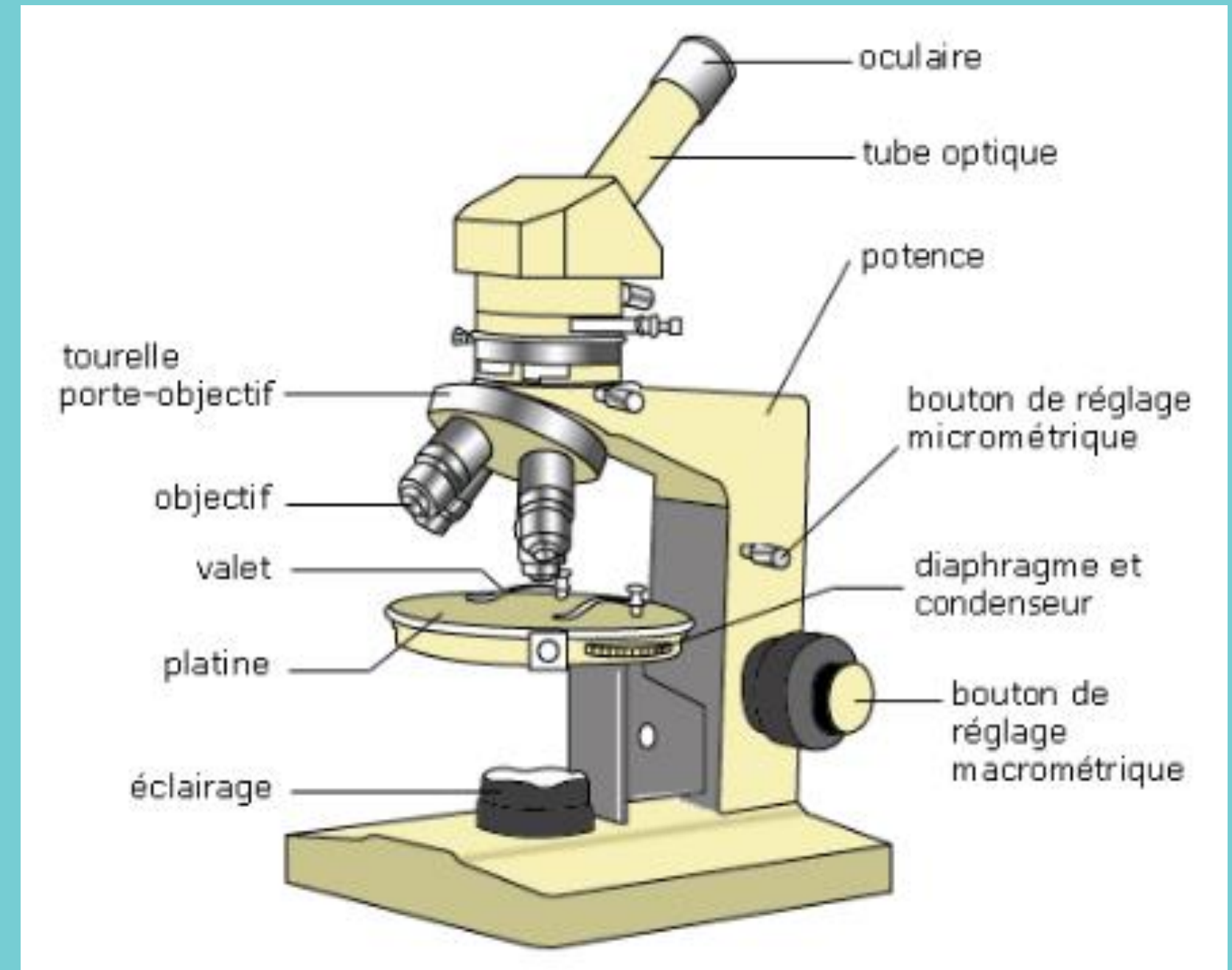


Figure 1 : Description du microscope optique (®[Prof Innovant](#))

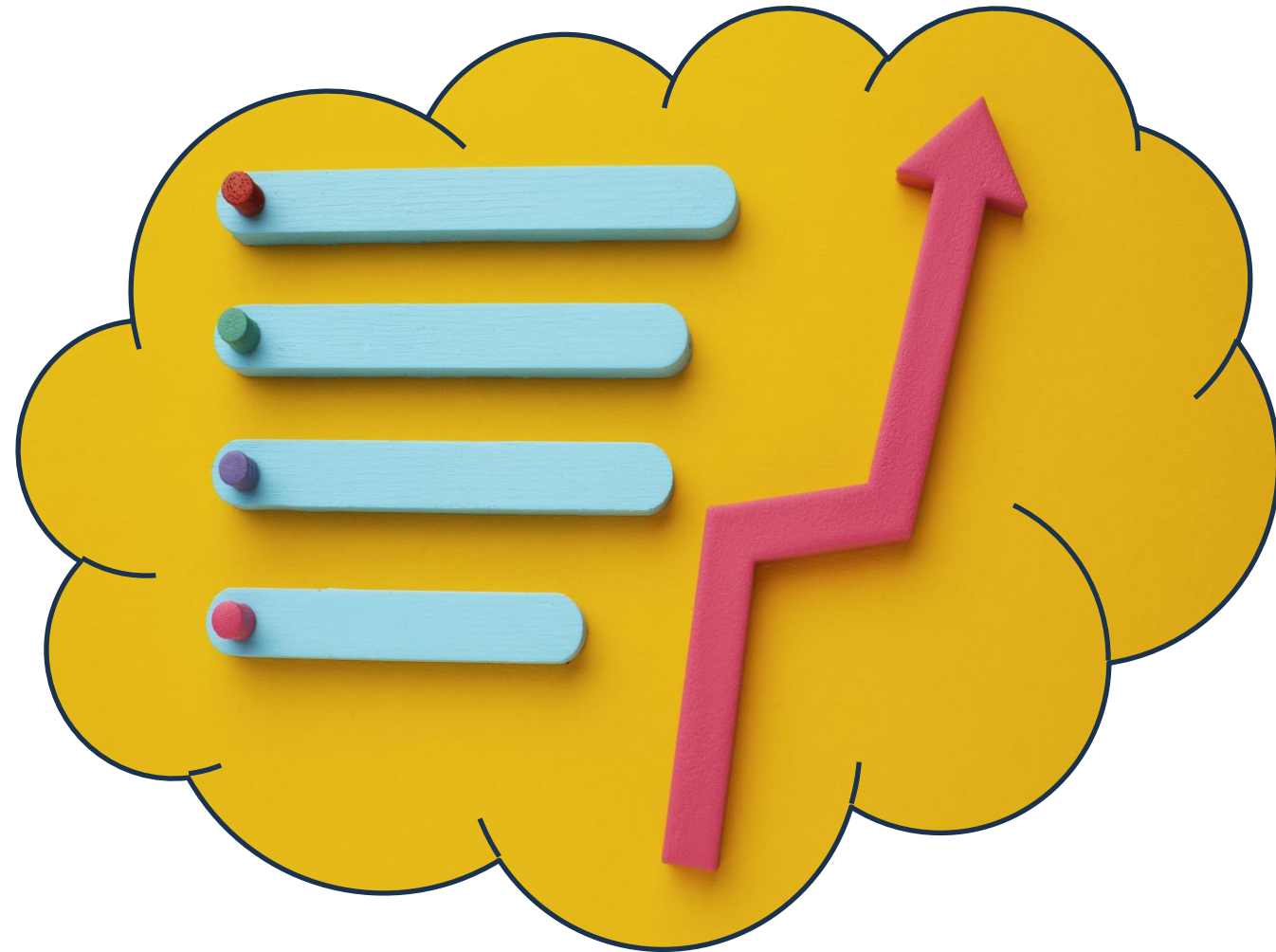
Reminding

Project Objective

- Design and build a low-cost, open-source optical microscope
- Based on the Open Flexure Microscope concept
- Use 3D printing + accessible electronics
- Educational and experimental purposes



Figure 2 : Open flexure microscope (®[OFM](#))



Progress Indicator ?

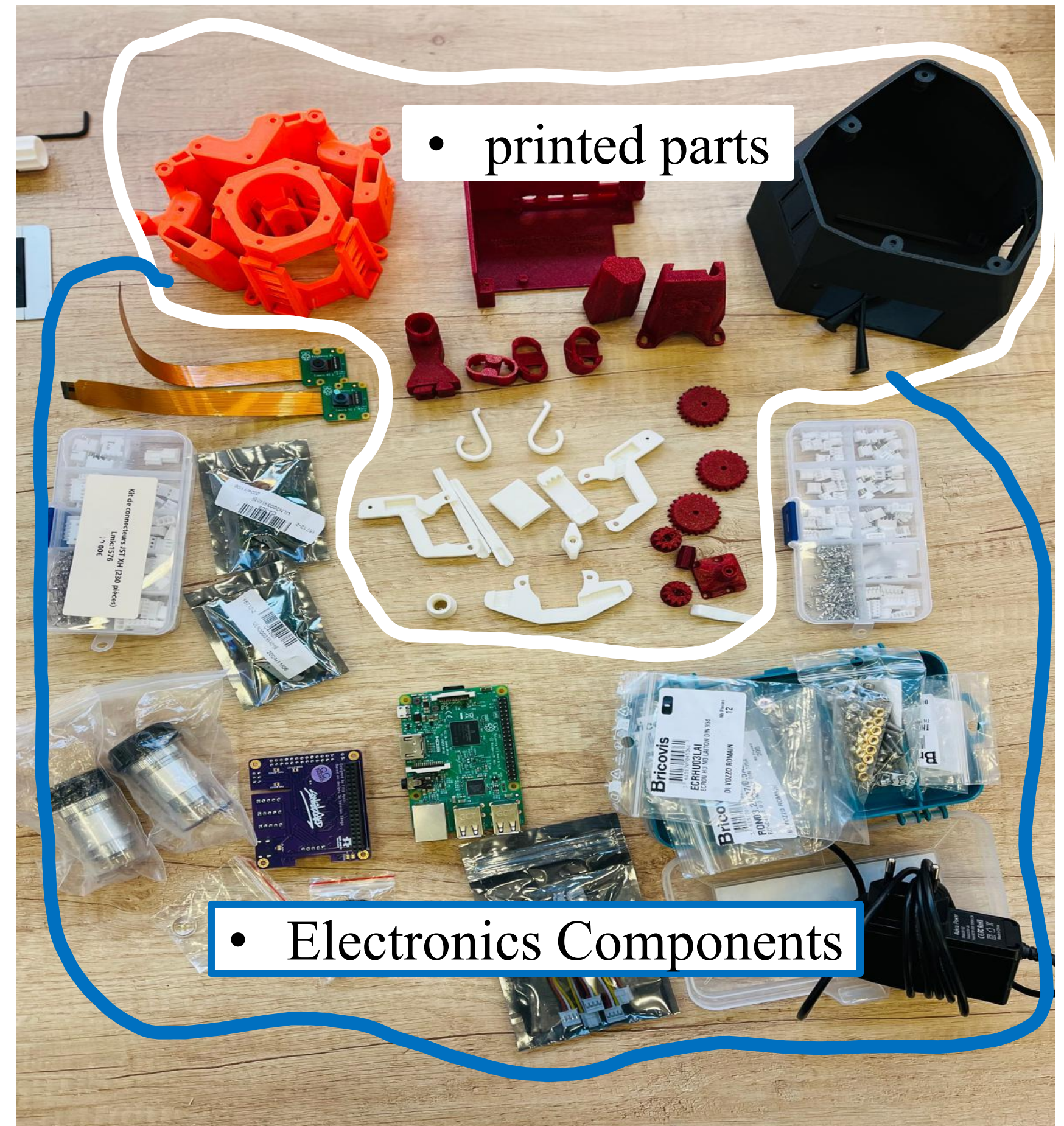
- Technical expertise
- Scientific and conceptual
- Project structuring and management
- Monitoring and traceability
- Equipment and logistics
- Transition to the implementation phase

	A	B	C	D	E	F	G	H	I
	Name in documentation	Category	Qty	Full Name	Notes	Supplier 1	Part No	Link	Supplier
1	0.5mm polypropylene	material	4 cm^2	White polypropylene sheet, 0.5mm thick		Kitronik	43142	https://kitronik.co.uk/collections/polypropylene-sheets/products/43142-white-polypropylene-sheet-05mm-x-1020mm-x-720mm	
2	2 pin Du Pont connector female housing	electronic	2	1x2 Crimp Connector Housing		Proto Pic	Search for Crimp Connector Housings	https://proto-pic.co.uk/product/pololu-6059-01-crimp-connector-housing-range-254mm/?attribute_pa_housing-size=1x2-25-pack	
3	200mm Pi Camera ribbon cable	electronic	1	200mm ribbon cable for Raspberry Pi Camera Module V2.1.	Make sure that the cable is Type B, as identified in the part page	Farnell (Part number also for Newark and Onecall)	3385343	https://uk.farnell.com/multi-omp-pro/mp-ffca10152003b/ffc-cord-15p-opposite-side-200mm/dp/3385343	The Pi
4	28BYJ-48 micro geared stepper motors	electronic	3	28BYJ-48 Stepper Motors					
5	condenser lens	optical	1	PMMA Condenser Lens		AliExpress	Search for '13mm 5mm PMMA lens'	https://www.aliexpress.com/wholesale?SearchText=13mm+5mm+pmma+lens	
6	extra M3 nut	tool	1	Hex Nut M3 in A2 Stainless	For mounting the hex bolt in the large gear	McMaster Carr (0)	Search for ISO 4032	https://www.mcmaster.com/ISO-4032	McMaster

- 3D printing phase (as things go along)
- Reception of electronic components (11/12/2025)

→ Includes components required for:

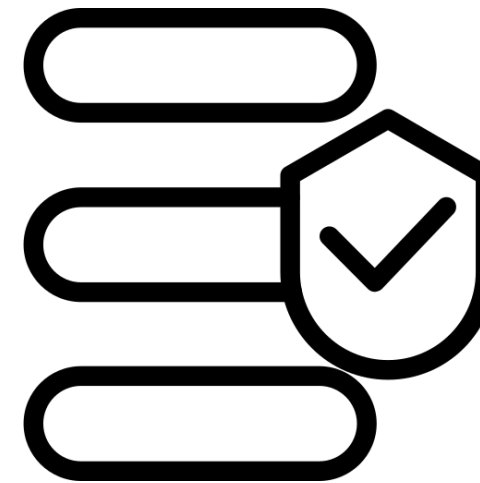
- **Illumination**
- **Cards**
- **Camera integration**



Indicator	Current status
3D printing autonomy	✓ Acquire
Scientific understanding	✓ Advanced course
Planning & organization	✓ Structured
Reporting & documentation	● In place
Composants & logistique	● Components & Logistics
Assembly	● Started



-  Folder with items to be printed
-  List of Bill of materials
-  Technical documentation on the steps of 3D printing, assembly and calibration.
-  Demonstration of use (e.g. image acquisition, calibration, maintenance, configuration of plugins).
-  Functional microscope (prototype)



Context

Goal

Progress Overview

Deliverables

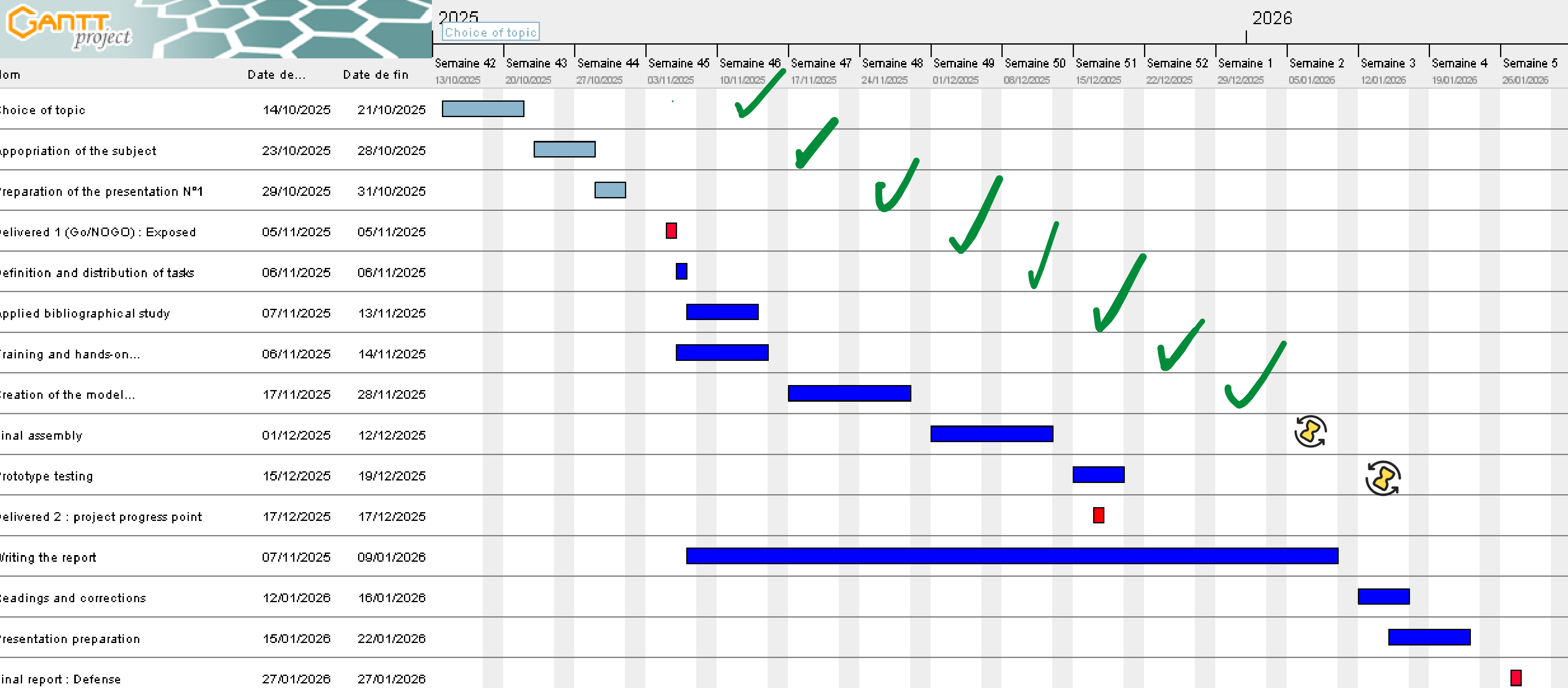
Risk

Planning

Conclusion

Risk	Gravity (1 à 4)	Probability of occurrence (1 à 4)	Met
Coordination, communication, organization	3	2	Yes
Incorrect identification of needs	3	2	Yes
None respect of schedule	1	2	Yes
Poor task allocation/prioritization	2	3	Yes
Technical difficulty	3	2	Yes





Context

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Redaction



Continue assembly



**Move forward with
programming**

Thanks

For

ATTENTION





MAIGA Yahiya Abdou

Hydrogeologist/ hydrologist,
Synthesizing & modeling

Motivation :

- develop skills in electronics and instrumentation
- design an instrument, open source and reproducible
- explore the application of the microscope



MUSHTAQ Aqib

Specialist of Internet
Of Things

Motivation :

- experiment with connectivity and data transmission
- develop a smart interface
- linking IoT concepts to experimental science,
- integrate sensors and electronic components